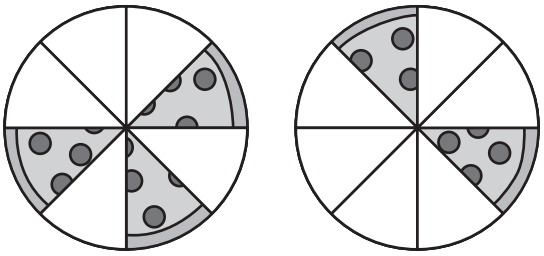
1 A family ordered two pizzas for dinner. Both pizzas were the same size. Each slice of pizza was the same size.

This diagram shows the amount of pizza remaining after dinner.



Which of these equations shows how to find the fraction of a whole pizza that was **remaining** after dinner?

Ⓐ $\frac{3}{5} + \frac{2}{6} = \frac{5}{11}$

Ⓑ $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$

Ⓒ $\frac{5}{8} + \frac{6}{8} = \frac{11}{8}$

Ⓓ $\frac{3}{8} + \frac{2}{8} = \frac{5}{16}$

2 Claire correctly worked out the answer to a subtraction problem. Which of these could be Claire's problem and work?

Ⓐ

$$\begin{array}{r} 3012 \\ -176 \\ \hline 3164 \end{array}$$

Ⓑ

$$\begin{array}{r} {\scriptstyle 2\ 9\ 10} \\ \cancel{3}\cancel{0}\cancel{1}{}^{1}2 \\ -176 \\ \hline 2836 \end{array}$$

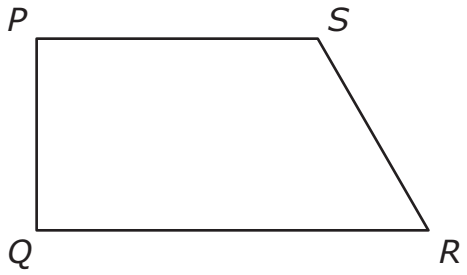
Ⓒ

$$\begin{array}{r} {\scriptstyle 9\ 10} \\ 3\cancel{0}\cancel{1}{}^{1}2 \\ -176 \\ \hline 3836 \end{array}$$

Ⓓ

$$\begin{array}{r} {\scriptstyle 2\ 9\ 9} \\ \cancel{3}\cancel{0}\cancel{1}{}^{1}2 \\ -176 \\ \hline 2826 \end{array}$$

3 A student created this figure by drawing line segments and angles.



Which line segments and angles did the student use to create the figure?

Select the **two** correct answers.

Ⓐ line segment *PR*

Ⓑ line segment *PQ*

Ⓒ line segment *QS*

Ⓓ angle *SPQ*

Ⓔ angle *SQR*

Ⓕ angle *QSR*

4 In which of these numbers does the 7 have a value that is **ten** times the value of the 7 in 9,176?

Ⓐ 2,473

Ⓑ 3,724

Ⓒ 4,327

Ⓓ 7,432

5 Which of these comparisons are correct?

Select the **two** correct answers.

Ⓐ $0.29 < 0.8$

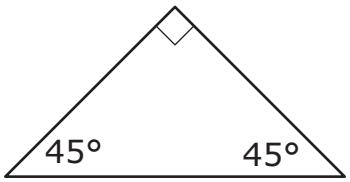
Ⓑ $0.29 > 0.8$

Ⓒ $0.29 = 0.8$

Ⓓ $0.8 < 0.29$

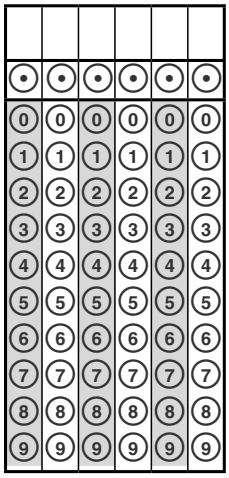
Ⓔ $0.8 > 0.29$

6 A triangle is shown.



How many lines of symmetry does the triangle have?

Enter your answer in the answer boxes at the top of the answer grid **and** completely fill the matching circles.



7 Which of these statements are correct?

Select the **three** correct answers.

Ⓐ $36 \div p = 4$ has a related multiplication fact of $p \times 4 = 36$.

Ⓑ $36 \div p = 4$ has a related multiplication fact of $36 \times p = 4$.

Ⓒ $s \div 7 = 5$ has a related multiplication fact of $s \times 5 = 7$.

Ⓓ $s \div 7 = 5$ has a related multiplication fact of $7 \times 5 = s$.

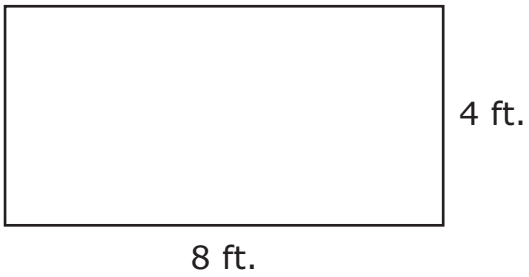
Ⓔ $72 \div 12 = a$ has a related multiplication fact of $a \times 12 = 72$.

Ⓕ $72 \div 12 = a$ has a related multiplication fact of $a \times 72 = 12$.

This question has four parts. Be sure to label each part of your response.

8 There is a garden, a patio, and a flower bed in the backyard of a house. The garden, the patio, and the flower bed are each in the shape of a rectangle.

A. The garden has a length of 4 feet and a width of 8 feet, as shown in this diagram.



What is the area, in square feet, of the garden?

B. The patio has a length of 5 feet and an area of 35 square feet.

What is the width, in feet, of the patio? Show or explain how you got your answer.

C. The owner of the house thinks the garden and the patio have the same perimeter.

Is the owner correct? Explain your reasoning.

D. The area of the flower bed is **less than** the area of the garden. The perimeter of the flower bed is **equal** to the perimeter of the patio.

What could be the length **and** the width of the flower bed? Explain how you know your answer is correct.

Write your answers on the next page.

9 A student drinks $\frac{2}{5}$ liter of water each morning.

Which of these is the total amount of water, in liters, the student drinks over 4 mornings?

Ⓐ $\frac{6}{5}$ liters

Ⓑ $\frac{8}{5}$ liters

Ⓒ $\frac{8}{10}$ liter

Ⓓ $\frac{8}{20}$ liter

10 A teacher has 8 sheets of stickers for decorating posters.

- Each sheet has 10 stickers.
- Each poster will have exactly 3 stickers.

What is the **greatest** number of posters the teacher can decorate?

Ⓐ 25 posters

Ⓑ 26 posters

Ⓒ 27 posters

Ⓓ 28 posters

This question has two parts.

11 On field day, a student recorded his finish times for three different races. He also recorded the distances of four of his softball throws.

Part A

The student's finish times for the three races are shown.

- 3 minutes
- 4 minutes, 39 seconds
- 10 minutes, 17 seconds

Which table shows the student's finish times, in **seconds**, for the three races?

Ⓐ **Finish Times**

Time	Time in Seconds
3 minutes	180
4 minutes, 39 seconds	179
10 minutes, 17 seconds	617

Ⓑ **Finish Times**

Time	Time in Seconds
3 minutes	180
4 minutes, 39 seconds	279
10 minutes, 17 seconds	617

Ⓒ **Finish Times**

Time	Time in Seconds
3 minutes	180
4 minutes, 39 seconds	279
10 minutes, 17 seconds	6,017

Ⓓ **Finish Times**

Time	Time in Seconds
3 minutes	300
4 minutes, 39 seconds	439
10 minutes, 17 seconds	1,017

Part B

The distances of the student's four softball throws were recorded using different units of measurement, as shown.

120 inches	3 yards	8 feet	100 inches

Which of these shows the distances of the softball throws in the correct order from **least to greatest**?

Ⓐ

8 feet	100 inches	3 yards	120 inches

Ⓑ

8 feet	3 yards	100 inches	120 inches

Ⓒ

100 inches	120 inches	8 feet	3 yards

Ⓓ

3 yards	8 feet	100 inches	120 inches

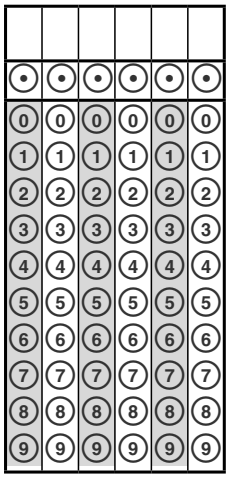
12 Lashawn knows this number sentence is true.

$$4 \times 7 = 28$$

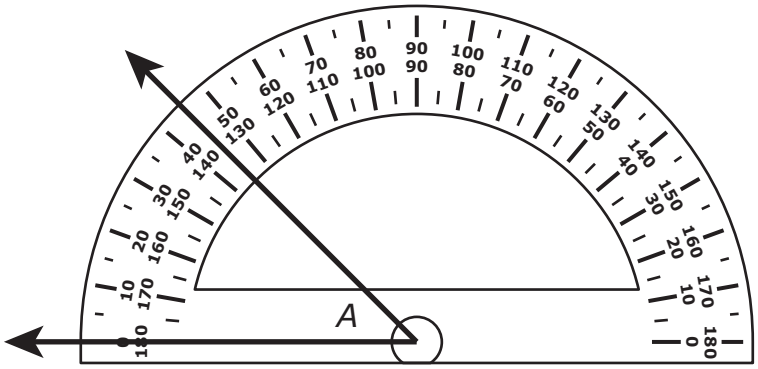
What is the value of $\boxed{?}$ that makes this number sentence true?

$$4 \times 700 = \boxed{?}$$

Enter your answer in the answer boxes at the top of the answer grid **and** completely fill the matching circles.

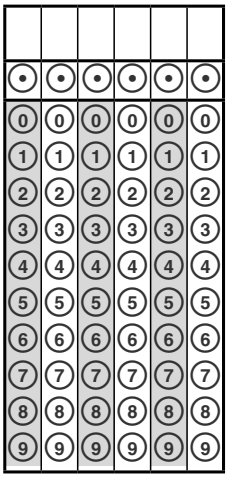


13 A protractor is used to measure angle *A*, as shown.



What is the measure, in degrees, of angle *A*?

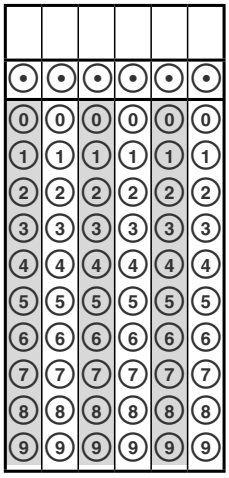
Enter your answer in the answer boxes at the top of the answer grid **and** completely fill the matching circles.



14 What value of c makes this equation true?

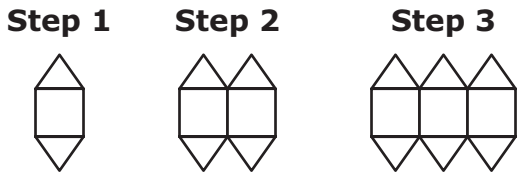
$$\frac{4}{6} = \frac{8}{c}$$

Enter your answer in the answer boxes at the top of the answer grid **and** completely fill the matching circles.



This question has four parts. Be sure to label each part of your response.

15 A student uses squares and triangles to make a pattern. In each step of the pattern, the student adds 1 square and 2 triangles, as shown.



The student continues the pattern.

A. What is the total number of triangles in Step 4 of the pattern?

B. What is the total number of **squares** in Step 6 of the pattern? Explain how you know your answer is correct.

C. What is the total number of **triangles** in Step 9 of the pattern? Explain how you can get your answer by using multiplication.

D. One step in the pattern will have a total of 64 triangles.

What is the total number of **squares** in that step? Show or explain how you got your answer.

Write your answers on the next page.

16 Which of these sentences are correct?

Select the **three** correct answers.

Ⓐ The number 142,839 rounded to the nearest **thousand** is 142,000.

Ⓑ The number 142,839 rounded to the nearest **thousand** is 143,000.

Ⓒ The number 142,839 rounded to the nearest **ten thousand** is 140,000.

Ⓓ The number 142,839 rounded to the nearest **ten thousand** is 150,000.

Ⓔ The number 142,839 rounded to the nearest **hundred thousand** is 100,000.

Ⓕ The number 142,839 rounded to the nearest **hundred thousand** is 200,000.

17 Which two fractions are equivalent?

Select the **two** correct answers.

Ⓐ $\frac{4}{1}$

Ⓑ $\frac{40}{1}$

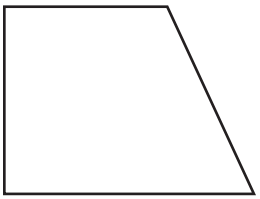
Ⓒ $\frac{4}{10}$

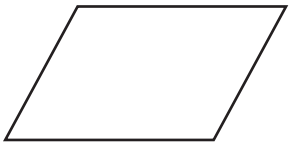
Ⓓ $\frac{4}{100}$

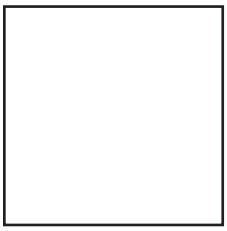
Ⓔ $\frac{40}{100}$

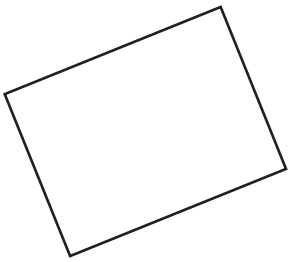
18 Which of these shapes appear to have at least one pair of parallel sides **and** at least one pair of perpendicular sides?

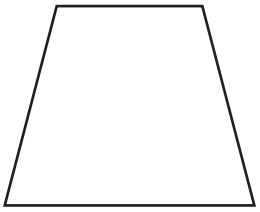
Select the **three** correct shapes.

Ⓐ 

Ⓑ 

Ⓒ 

Ⓓ 

Ⓔ 

19 Which of these fractions is equivalent to 0.3?

Ⓐ $\frac{1}{3}$

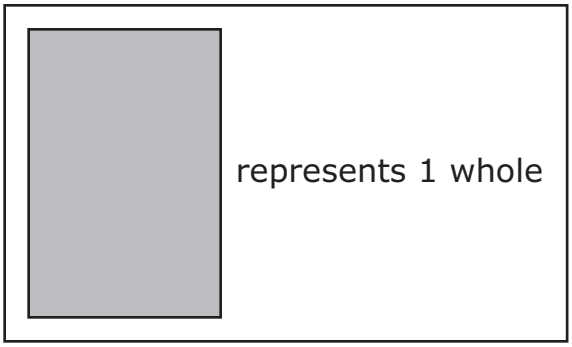
Ⓑ $\frac{1}{30}$

Ⓒ $\frac{3}{10}$

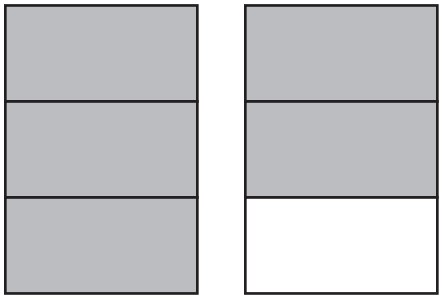
Ⓓ $\frac{3}{100}$

20 Which fraction model is shaded to show the product of this expression?

$$4 \times \frac{1}{3}$$



Ⓐ



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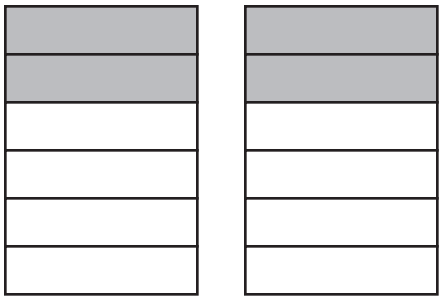

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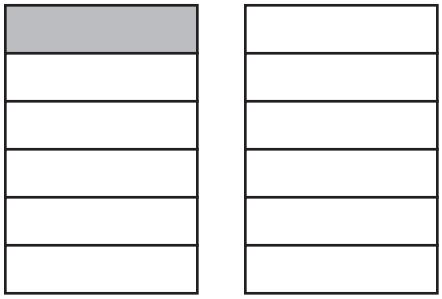
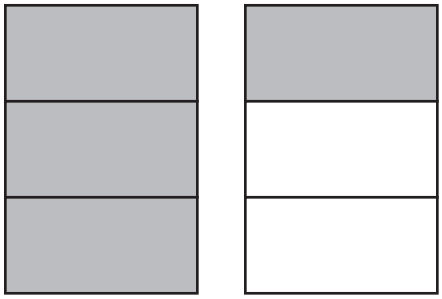
Ⓓ